Historical density overlap instruments

Controls: Match of Historical to Modern Geographic Units; Max Density Overlap; Weighted Density Overlap; Minimum Overlap (%).

1. Start with historical population density. Let j identify a historical county in census year t. Its population is P_{jt} and its mapped area is a_{jt}, in square kilometres. Its population density is P_{jt}/a_{jt}. Let $B_{jg,t}$ be the area shared with the modern observation g, which can be a county or an MSA. The overlap methods use historical counties with positive population density in the selected historical scope.

2. The overlap threshold selects eligible counties. For an MSA, a historical county qualifies when $B_{jg,t}/a_{jt}$ is at least the chosen threshold. For a modern county, it qualifies when $B_{jg,t}/A_g$ exceeds the threshold, where A_g is the modern county's mapped area. The denominator therefore differs between the two analysis levels. At 5%, the MSA rule asks whether at least 5% of the historical county lies inside the MSA; the County rule asks whether the historical county covers more than 5% of the modern county.

3. Maximum density selects one historical county. Write $\mathcal{J}_{g,t}$ for the eligible historical counties and choose

$$j^* = \arg\max_{j \in \mathcal{J}_{g,t}} \frac{P_{jt}}{a_{jt}}.$$

Under the *Employment density* approach, the instrument is $z_{g,t} = P_{j^*t}/a_{j^*t}$. Under *Employment*, it is the full population P_{j^*t}, expressed in thousands under Levels. Selection is still based on density. No fraction of that county's population is allocated to the modern unit. If no county qualifies, the instrument is unavailable.

4. Weighted density uses all eligible overlaps. At County level, *Weighted Density Overlap* instead uses

$$z_{g,t} = \sum_{j \in \mathcal{J}_{g,t}} \frac{B_{jg,t}}{\sum_{k \in \mathcal{J}_{g,t}} B_{kg,t}} \frac{P_{jt}}{a_{jt}}.$$

This is an average of historical densities, weighted by shared area. It is available only for the employment-density approach and is distinct from the area-weighted population total. Raising the threshold can change the selected historical county, the weighted average and sample availability. Consequently, threshold comparisons can reflect changes in both the instrument and the sample. The scale control can apply a natural logarithm after construction.